

Devin Autonomous Issue Solver

Turning a stale GitHub backlog into reviewed PRs — automatically.

Take-home Demo for Cognition

What is Devin?

An autonomous software engineer

- Works in its own containerized cloud environment
- Clones repo · reads code · edits · **runs tests in a real shell** · opens a PR
- You delegate a *whole ticket*, not a line of autocomplete code suggestions

The cost of issues backlog

Time solving + context switch.

- **For each issue, an engineer must:**
 - Remember context, setup environment, diagnose, debug, fix, write tests
- **The Toll:**
 - Dropping deep work, wasted focus and time.
 - Lowered morale

System Architecture

FastAPI app polls inside docker.

Today (demo)	In production
Polling every 3 min	GitHub webhook (instant)
JSON-file store	Postgres (multi-replica)
Token-free public reads	Service account / app token
Server-rendered HTML	Decoupled Next.js frontend

- **Other** structured output schema → root cause, test command, test stdout. Diffs from Github PR

Why Devin?

	Human Flow	In-Editor Assistant	Devin (This System)
Workspace Setup	every time	every time	none
Who Drives	engineer	engineer	autonomous
Context Switch	yes	yes	no — reviews finished PR
Parallel Issues	no	no	yes (multiple sessions)

- **Trust Through Transparency:** Shows root cause analysis and raw test outputs. Human owns the merge.
- **Integrations:** Acts as a private bug fixer for non-technical users (through Slack/Jira for example).

